MODULE 01 · DOMAIN 1 - SNOWFLAKE AI DATA CLOUD FEATURES AND ARCHITECTURE

Three layers, and why everything else follows

The one architectural decision that explains cloning, sharing, elastic scale and per-second billing.

DOMAIN 1 of 5 WEIGHT 31% MODULE 1 of 31

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In a traditional warehouse the data lives on the machine that queries it. Snowflake broke that link.

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Snowflake is a self-managed service

What the documentation means by that, and what it rules out

- **No hardware** to select, install, configure or manage. Virtual or physical.
- **Virtually no software** to install or manage. Maintenance, upgrades and tuning are Snowflake's job.
- You **cannot** run Snowflake locally, on-premises, or on private cloud infrastructure.
- It runs on public cloud infrastructure: **AWS, Azure or Google Cloud**. You choose the platform and region.

> ON THE EXAM
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> The last two points are a question waiting to happen. 'Deploy Snowflake in our own data centre' is never the answer.

The three layers, bottom to top

01

Database storage

Data, reorganised into compressed columnar micro-partitions. Snowflake manages all of it.

02

Compute

Virtual warehouses. Independent MPP clusters that read the storage layer.

03

Cloud services

The coordinator. Authentication, metadata, optimisation, access control.

ON THE EXAM

Learn these names exactly. 'Which layer is responsible for X' is one of the most reliable question formats on the paper.

What actually lives in each layer

LAYER	RESPONSIBILITIES
Storage	Your table data. Micro-partitions, compression, file size, structure, statistics. Not directly visible to you.
Compute	Query execution. Loading, DML, Snowpark code in Java, Python and Scala.
Cloud services	Authentication and access control. Metadata. Query parsing and optimisation. Infrastructure management. Horizon Catalog. Compliance.

ON THE EXAM

The result cache and the metadata cache both live in cloud services. That is why some queries return with no warehouse running.

Where the design came from

SHARED-DISK

One central store, many nodes

- All nodes reach the same data
- Simple to manage: one copy of the truth
- Nodes contend for the same store
- Scaling hits a ceiling

SHARED-NOTHING

Each node owns its slice

- Data partitioned across nodes
- Massively parallel, scales out well
- Adding a node means reshuffling data
- Storage and compute must grow together

HOW SNOWFLAKE RESOLVES IT

Snowflake is a hybrid: the management simplicity of shared-disk with the scale-out performance of shared-nothing.

Every feature you will study in domains two to five is a **receipt** for this one decision.

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Four features that are really one idea

- **Zero-copy cloning.** A clone points at existing micro-partitions. No data is duplicated, so no storage is charged until it diverges.
- **Secure Data Sharing.** No data is copied to the consumer. They read your storage using their own compute.
- **Workload isolation.** Each warehouse is independent and has no effect on the performance of any other.
- **Elastic, per-second compute.** A warehouse holds no data, so suspending it loses nothing but the local cache.

> ON THE EXAM
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> When a question describes one of these, the reasoning behind the right answer is always separation of storage and compute.

Three meters, billed three different ways

WHAT YOU PAY FOR	HOW IT IS CHARGED
Virtual warehouse compute	Credits, per second, 60-second minimum on each start or resume.
Storage	Flat monthly rate per compressed terabyte. Varies by provider and region.
Cloud services	Only the portion exceeding 10% of that day's warehouse credits.
Serverless compute	Each feature's own rate. Snowpipe, clustering, search optimization, tasks.
Data transfer	Egress across regions or clouds. Ingress is free.

ON THE EXAM

Separate meters. A cost question is usually asking which one an activity lands on.

The cloud services allowance, precisely

- Cloud services is charged **only if** the day's cloud services credits exceed **10% of that day's warehouse credits**.
- Calculated **daily, in UTC**. The monthly figure is the sum of those daily calculations.
- The adjustment never exceeds actual usage. If you used less than the allowance, you pay nothing for it.
- **Serverless compute does not count** towards the 10% calculation.
- Resource monitors **ignore** the adjustment and count all cloud services credits, even unbilled ones.

> ON THE EXAM
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> Metadata-heavy patterns push you over: thousands of tiny queries, or heavy INFORMATION_SCHEMA use with little warehouse work.

What the storage layer will hold

KIND	MEANING
Structured	Rows and columns. A strict tabular schema.
Semi-structured	JSON, XML, Avro, ORC, Parquet. A flexible schema.
Unstructured	Documents, images, audio. No inherent schema.

ON THE EXAM

All three, in one platform, queried with ANSI SQL. Semi-structured lands in a VARIANT column; unstructured is reached through stages and the FILE type.

Three families of table

FAMILY	WHERE THE DATA SITS, AND WHAT IT IS FOR
Snowflake tables	Snowflake's own storage, in micro-partitions. The default. Ideal for data warehousing.
Apache Iceberg tables	External cloud storage that you manage. For data lakes you cannot or will not move into Snowflake.
Hybrid tables	Optimised for low-latency reads and writes. Row locking, unique and referential constraints. Unistore.

ON THE EXAM

Iceberg and hybrid tables are new to COF-C03. Recognising which family a scenario needs is enough at this stage.

Snowgrid, and why regions matter

- An account lives in **one cloud provider and one region**. You choose both when it is created.
- **Snowgrid** is the cross-region, cross-cloud layer that connects accounts across that boundary.
- It underpins **listings** for collaboration across clouds, consistent policy across regions, and **replication** for disaster recovery.
- Region matters. A direct share does not cross regions on its own. Domain five deals with what does.

> ON THE EXAM
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> When a scenario names two regions or two cloud providers, that detail was planted deliberately. It usually rules out the obvious answer.

How this module gets tested

- **'Which layer handles X.'** Authentication, metadata, optimisation and access control are all cloud services.
- **'Why is this billed.'** Work out which of the meters the activity lands on.
- **'Two teams interfering.'** Separate warehouses. Almost never a bigger single warehouse.
- **Anything on-premises.** Not possible. Distrust the option immediately.

> ON THE EXAM
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> Four question shapes, all answerable from one diagram. This is why architecture is worth 31%.

An analytics team runs interactive dashboards during the working day. A data engineering team runs a heavy nightly transformation that has begun overrunning into the morning.

When the two overlap, dashboard queries that normally return in two seconds take over a minute. Both teams currently use the same virtual warehouse. Budget is under scrutiny.

What is the most appropriate remedy?

A. Resize the shared warehouse from Medium to 2X-Large so it can absorb both workloads.

B. Create a second virtual warehouse and point the transformation job at it.

C. Move the dashboard tables into a separate database so the two workloads read different storage.

D. Enable multi-cluster scaling on the shared warehouse and set the scaling policy to Economy.

◆ PAUSE AND DECIDE

What is the most appropriate remedy?

A. Resize the shared warehouse from Medium to 2X-Large.

Would help, and costs sixteen times as much per hour while the two workloads still contend. Budget was named as a constraint.

B. Create a second virtual warehouse for the transformation job.

Warehouses are independent and have no effect on each other. The workloads stop competing, and each is sized for its own job.

C. Move the dashboard tables into a separate database.

Databases are a logical grouping. Contention was for compute, and storage was never the bottleneck.

D. Multi-cluster with an Economy scaling policy.

Multi-cluster answers concurrency, not two workloads of different shapes. Economy also makes it slower to add a cluster.

THE SIGNAL

'Budget is under scrutiny' eliminated A. Read those sentences as instructions, not background.

> An analyst runs a query at nine in the morning. A colleague runs the identical query twenty minutes later against unchanged data, and it returns in well under a second.
>
> Checking Snowsight, the colleague finds that no virtual warehouse was running at the time, and no warehouse credits were consumed by their query.

Which layer served the second query?

A. The compute layer, using a warehouse that auto-resumed and suspended too quickly to appear.

B. The database storage layer, which serves cached micro-partitions directly.

C. The cloud services layer, from the result cache.

D. The compute layer, from the warehouse local disk cache retained after suspension.

◆ PAUSE AND DECIDE

Which layer served the second query?

A Compute, from a warehouse that resumed and suspended quickly.

A resume would have consumed the 60-second minimum. The scenario states no warehouse credits were used.

B Storage, serving cached micro-partitions directly.

The storage layer holds data. It does not execute or serve queries on its own.

C Cloud services, from the result cache.

The result cache lives in cloud services, which is precisely why it can serve a query with every warehouse suspended.

D Compute, from the warehouse local disk cache.

That cache is on the warehouse's local storage and is lost when the warehouse suspends.

THE SIGNAL

'No warehouse credits were consumed' is the whole question. Only one layer can serve a result without compute.

A finance analyst is reviewing a monthly statement. On one day the account consumed 400 virtual warehouse credits and 32 cloud services credits.

On another day it consumed 50 warehouse credits and 30 cloud services credits. The analyst wants to understand why the second day contributed a cloud services charge and the first did not.

Which explanation is correct?

A Cloud services is charged only above 10% of that day's warehouse credits, calculated daily.

B Cloud services is free below a fixed monthly ceiling, and the second day happened to cross it.

C Cloud services is charged whenever serverless features run, and serverless ran only on the second day.

D Cloud services is free on days when a resource monitor is attached to the account.

◆ PAUSE AND DECIDE

Which explanation is correct?

A Charged only above 10% of that day's warehouse credits, daily.

Day one: 10% of 400 is 40, and 32 is under it. Day two: 10% of 50 is 5, so 25 credits are billable.

B A fixed monthly ceiling that the second day crossed.

The allowance is proportional to warehouse usage and recalculated every day, not a fixed monthly figure.

C Charged whenever serverless features run.

Backwards. Serverless is billed separately and is explicitly excluded from the 10% calculation.

D Free when a resource monitor is attached.

Resource monitors observe and cap credit usage. They never change what is billable.

THE SIGNAL

A low-warehouse, high-metadata day is the expensive one. The allowance shrinks with the warehouse usage it is proportional to.

> A public sector organisation has a policy requiring that analytical data be processed on infrastructure it controls. Its architects want to keep using Snowflake and propose installing it in the organisation's own data centre, alongside its existing systems.
>
> A second requirement states that compute resources must not be shared with other Snowflake customers.

How should an architect respond?

A Deploy Snowflake on-premises using the private cloud installer, then enable Tri-Secret Secure.

B Use a reader account, which runs inside the organisation's own cloud subscription.

C Deploy a Business Critical account and add PrivateLink, which makes the deployment on-premises.

D An on-premises deployment is not possible. Virtual Private Snowflake gives a dedicated, isolated environment instead.

◆ PAUSE AND DECIDE

How should an architect respond?

A. Deploy on-premises with a private cloud installer.

No such installer exists. The documentation states you cannot run Snowflake locally or on private cloud infrastructure.

B. A reader account in the organisation's own subscription.

A reader account is provisioned and owned by a sharing provider. It is not customer-hosted infrastructure.

C. Business Critical plus PrivateLink.

PrivateLink gives private network connectivity. It does not move where Snowflake runs.

D. Not possible on-premises. Virtual Private Snowflake gives isolation instead.

Correctly refuses the impossible half and answers the real requirement, which was isolation rather than location.

THE SIGNAL

Two requirements, and only one is satisfiable. Saying so is the answer, not a failure to find one.

What to carry into the next module

- **Three layers.** Storage holds. Compute executes. Cloud services coordinates, and holds the metadata and result caches.
- **Hybrid architecture.** Central repository from shared-disk, MPP scale-out from shared-nothing.
- **Separation is the reason** cloning is free, sharing copies nothing, warehouses cannot interfere, and billing is per second.
- **Cloud services allowance:** 10% of the day's warehouse credits, daily, in UTC. Serverless excluded.
- **Never on-premises.** Public cloud only, on one provider and one region per account.

> ON THE EXAM
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> Next: editions. Not a topic so much as a question format, because 'what is the MINIMUM edition' is guaranteed to appear.